

Smart Learn: An Adaptive Mobile Platform for Neurodevelopmental Learning Disorders

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Project Proposal Report

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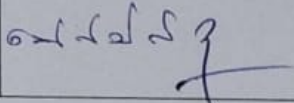
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Developmental Dyscalculia is a specific learning disorder that affects a child's ability to understand numbers, perform arithmetic operations, and develop foundational number sense. In the Sri Lankan primary education context, children experiencing dyscalculia-related learning difficulties often struggle with essential numeracy skills such as number recognition, quantity comparison, place value understanding, and basic arithmetic reasoning. These challenges frequently lead to persistent academic difficulties in mathematics and reduced confidence during the early stages of schooling.

Despite the increasing use of digital learning technologies in education, the availability of localized digital tools designed specifically to support dyscalculia-related learning difficulties among Sri Lankan primary school learners remains limited. Most existing mathematics learning applications focus on general arithmetic practice and are primarily developed for English-speaking learning environments. Consequently, these systems do not adequately address the cognitive challenges associated with dyscalculia or provide personalized learning interventions tailored to the Sri Lankan educational context.

This research identifies a gap in accessible digital learning platforms that integrate assessment, adaptive learning, and continuous progress monitoring within a single system. To address this gap, the proposed system, SmartLearn+, is an adaptive mobile-based learning platform designed to assess learners' numeracy abilities and dynamically personalize mathematical learning activities.

A key novelty of the proposed dyscalculia module is the integration of a multisensory learning approach, which combines visual object-based number representation, Sinhala audio guidance, and interactive touch-based game activities. This approach enables children to associate numerical concepts through multiple sensory channels, improving number recognition, quantity understanding, and basic arithmetic reasoning. By engaging visual, auditory, and interactive learning modalities simultaneously, the system aims to provide a more effective learning experience for children with dyscalculia.

The platform will be developed using React Native for cross-platform mobile development, Node.js with Express for backend services, and MongoDB with Firebase for data management and real-time synchronization. A rule-based adaptive learning engine supported by learning analytics techniques will analyze learner performance indicators such as accuracy, response time, and error patterns to dynamically adjust the difficulty of learning activities.

Expected outcomes include measurable improvements in foundational numeracy performance, increased learner engagement through interactive educational activities, and improved identification of mathematical learning difficulties. The proposed system aims to promote inclusive digital learning within the Sri Lankan educational environment by providing accessible support for children experiencing dyscalculia-related challenges.

Keywords: Dyscalculia, Multisensory Learning, Adaptive Learning, Learning Analytics, Educational Technology.

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TABLE OF CONTENT

DECLARATION	2
ABSTRACT	3
ACKNOWLEDGEMENT	4
TABLE OF CONTENT	5
LIST OF TABLES	6
LIST OF FIGURES.....	7
LIST OF ABBREVIATIONS.....	8
LIST OF APPENDICES	9
1.INTRODUCTION	10
1.1 Background and Context.....	10
1.2 Literature Review.....	12
1.3 Research Gap	14
2. OBJECTIVE	16
2.1 Main Objective	16
2.2 Specific Objectives	16
3. METHODOLOGY.....	17
Research approach.....	17
Technical methods and algorithms	17
Tools and platforms with justification.....	19
Data collection and ethical considerations	20
Validation strategy and performance metrics	20
4.HIGH-LEVEL SYSTEM ARCHITECTURE.....	21
5.USER REQUIREMENTS.....	23
Stakeholder identification.....	23
Functional requirements	23
Non-functional requirements.....	24
6. COMMERCIALIZATION PLAN	25
7.BUDGET AND JUSTIFICATION	31
7.WORK BREAKDOWN STRUCTURE (WBS).....	32
8. GANTT CHART	33
10 REFERENCES LIST	34
11. APPENDICES.....	36

LIST OF TABLES

Table 01 - System Comparison	15
Table 02 - value proposition.....	25
Table 03 - Cost Estimation.....	26
Table 04 - Budget and Justification.....	27

LIST OF FIGURES

Figure 01 - System diagram.....	21
Figure 02- Component diagram for dysgraphia intervention module.....	21

LIST OF ABBREVIATIONS

Abbreviation	Word/Description
NLDs	Neurodevelopmental learning disorders
SDGs	Sustainable Development Goals
AR	Augmented Reality
DSR	Design Science Research
UI	User Interface
VS Code	Visual Studio Code
IDE	integrated development environment
IP	Intellectual Property
DD	Developmental Dyscalculia
AI	Artificial Intelligence
SVM	Support Vector Machine

LIST OF APPENDICES

1. Survey instruments.....	31
2. Risk analysis.....	32
3. Plagiarism Report.....	33

1.INTRODUCTION

1.1 Background and Context

Neurodevelopmental learning disorders (NLDs) affect cognitive processes related to learning, including reading, writing, mathematics, and memory. Common NLDs include dyslexia, dyscalculia, dysgraphia, and attention-related learning difficulties [1][2]. These conditions can significantly influence a child's ability to acquire fundamental academic skills and may persist into adulthood if appropriate intervention is not provided during early developmental stages. Developmental dyscalculia, a specific learning disorder, affects a child's ability to understand numbers, perform arithmetic operations, and develop numerical reasoning skills. Children with dyscalculia commonly experience difficulties with number recognition, counting, place value understanding, and basic mathematical calculations, which often result in long-term academic challenges and reduced confidence in mathematics [5].

In the Sri Lankan primary education context, mathematical learning difficulties among young learners are frequently observed during the early stages of schooling. However, identifying dyscalculia-related difficulties remains challenging due to limited awareness, lack of specialized assessment tools, and insufficient access to educational psychology services. As a result, many children experiencing persistent mathematical learning difficulties are often misinterpreted as having low academic ability rather than a specific learning disorder, which delays appropriate intervention and negatively affects their long-term academic development [3].

Traditional approaches for identifying and supporting dyscalculia typically rely on clinical psychological assessments, standardized cognitive tests, and specialist educational evaluations. Although these approaches can provide accurate diagnoses, they are often expensive, time-consuming, and inaccessible for many families. Furthermore, the availability of trained educational psychologists and special education professionals is limited, particularly in rural and underserved regions. Consequently, many children experiencing dyscalculia-related learning difficulties do not receive the individualized educational support required during the critical stages of early cognitive development [4].

Children between the ages of 6 and 10 years represent the most affected group, as this developmental stage is crucial for building foundational numeracy skills such as counting, number recognition, and basic arithmetic reasoning [4]. Early childhood is widely recognized as a sensitive period for cognitive and academic development, and difficulties encountered during this stage can produce long-term educational consequences. Research indicates that students with untreated dyscalculia frequently develop mathematics anxiety, reduced academic confidence, and lower participation in science, technology, engineering, and mathematics (STEM) disciplines later in life [5]. Therefore, early identification and targeted intervention are essential to minimize long-term educational and socio-economic impacts.

In recent years, various digital educational tools and mobile learning applications have been developed to assist children in practicing mathematical skills. These include arithmetic training applications, gamified learning platforms, and interactive educational software designed to

improve number recognition and problem-solving abilities [5]. However, many of these applications focus on general mathematics practice rather than addressing the specific cognitive challenges associated with dyscalculia. Most existing systems provide static exercises without adapting to individual learner difficulties or analyzing performance patterns.

Furthermore, many digital learning tools lack structured diagnostic screening, adaptive learning mechanisms, and detailed performance analytics that can assist parents and teachers in monitoring a child's progress. As a result, these systems provide limited support for personalized intervention and early identification of dyscalculia-related learning difficulties.

To address these challenges, this research proposes a game-based adaptive mobile learning system designed to support children with dyscalculia aged 6-8 years. The proposed system integrates multisensory learning techniques that combine visual object representation, Sinhala audio guidance, and interactive touch-based activities to enhance numerical understanding. By engaging multiple sensory channels simultaneously, the system aims to improve number recognition, counting ability, and basic mathematical reasoning among young learners.

The proposed system consists of several key components:

1. **Initial Skill Assessment Module** - The system evaluates a child's numeracy abilities using interactive game-based numeracy assessment tasks that evaluate children's mathematical performance using indicators such as response accuracy, response time, and task completion patterns. Performance indicators such as response accuracy and response time are used to categorize the learner's difficulty level (e.g., mild, moderate, or severe).
2. **Personalized Learning Path Recommendation** - Based on the learner's performance profile, the system recommends targeted learning activities designed to strengthen specific mathematical skills including number recognition, counting ability, and basic arithmetic reasoning.
3. **Adaptive Learning Mechanism** - The system dynamically adjusts task difficulty according to learner performance indicators such as accuracy, response time, and task completion rate. This ensures that learning activities remain appropriately challenging and supportive.
4. **Progress Monitoring and Performance Analytics** - The system continuously records learner performance data and generates progress reports through an analytics dashboard accessible to parents and teachers. After a predefined intervention period, the system reassesses the learner to evaluate improvement and adjust the learning pathway accordingly.

This research contributes to several United Nations Sustainable Development Goals (SDGs). Specifically, it supports SDG 4 (Quality Education) by promoting inclusive and equitable access to learning opportunities for children with learning difficulties. It also contributes to SDG 3 (Good Health and Well-Being) by supporting cognitive development and reducing learning-related academic stress [6]. In addition, the proposed digital intervention aligns

with SDG 10 (Reduced Inequalities) by providing accessible educational support tools for children in underserved communities.

From a research perspective, the proposed system aligns with the FoC Research Cluster: Technology Integration and Management, integrating artificial intelligence concepts with educational technology and learning analytics. By combining adaptive learning algorithms, multisensory interaction, and gamified educational activities, the system aims to deliver a scalable, intelligent, and accessible intervention platform capable of improving foundational numeracy skills among children with dyscalculia.

1.2 Literature Review

Digital technologies have increasingly been explored as potential tools for supporting children with mathematical learning disabilities, particularly developmental dyscalculia. Dyscalculia is characterized by persistent difficulties in understanding numerical concepts, performing arithmetic operations, and developing number sense. These challenges often manifest as difficulties in recognizing numerical magnitudes, understanding place value, memorizing arithmetic facts, and performing mental calculations.

Butterworth et al. examined the cognitive foundations of dyscalculia and identified that children with this disorder frequently demonstrate deficits in numerical magnitude processing and working memory associated with arithmetic tasks [7]. Their findings emphasize the importance of early intervention strategies designed to strengthen number sense and conceptual understanding rather than relying solely on repetitive arithmetic practice.

Geary investigated the neurocognitive mechanisms underlying mathematical learning disabilities and identified several contributing factors, including working memory limitations, processing speed deficits, and difficulties in numerical representation [8]. The study highlighted that targeted interventions focusing on conceptual number understanding can significantly improve mathematical learning outcomes among affected learners.

Several digital learning tools have been developed to support early numeracy development through interactive activities and gamified exercises. Educational applications such as DragonBox Numbers and Mathletics incorporate game-based learning mechanics to improve engagement and encourage repeated practice of mathematical skills. Studies evaluating these platforms have reported improvements in learner motivation and engagement in mathematical learning activities [9]. However, these systems primarily focus on general mathematics practice rather than addressing the specific cognitive deficits associated with dyscalculia.

Adaptive learning technologies have also been investigated as potential solutions for personalized education. Pane et al. analyzed personalized learning systems implemented in primary education and found that adaptive learning platforms can significantly improve student learning outcomes when instructional activities are tailored to individual performance levels [10]. Nevertheless, many existing adaptive learning platforms are designed for broad academic subjects and lack specialized modules that address dyscalculia-specific learning challenges.

Recent research has also explored the application of learning analytics and artificial intelligence techniques to analyze student learning behaviors. Ifenthaler and Yau highlighted

the role of learning analytics in identifying patterns of student performance and providing personalized recommendations in digital learning environments [11]. Their research indicates that analytics-driven educational systems can support the early identification of learning difficulties and enable data-driven intervention strategies.

In addition, machine learning techniques have been investigated for detecting mathematical learning disorders. Several studies have applied classification models to analyze student performance data obtained from arithmetic tasks in order to predict the likelihood of dyscalculia among learners [12]. Although these approaches demonstrate promising accuracy in identifying at-risk students, many of these systems function primarily as diagnostic tools rather than comprehensive intervention platforms designed to improve learning outcomes.

Kariyawasam et al. proposed a gamified mobile application for screening and intervention of learning disabilities, including dyslexia, dysgraphia, and dyscalculia [13]. For dyscalculia detection, the system evaluates children's abilities in counting, number comparison, and basic arithmetic using task-based activities. Performance indicators such as accuracy and response time are used as inputs to a Support Vector Machine (SVM) classifier to identify children with dyscalculia. If detected, the system provides game-based intervention activities to improve mathematical skills. However, the system mainly focuses on basic screening does not incorporate adaptive learning mechanisms or detailed performance analytics for personalized learning support.

Hewapathirana et al. developed the "Nana Shilpa" mobile system to screen and provide intervention for dyscalculia among Sri Lankan primary school students [14]. The system uses gamified activities and machine learning algorithms (SVM, Random Forest) to classify different types of dyscalculia and provides interactive games to support mathematical skills. While the system achieved high classification accuracy, it primarily focuses on screening and classification, offering limited adaptive learning or personalized numeracy interventions for individual students.

Overall, the existing literature demonstrates that digital learning technologies can improve engagement and support skill development among learners with mathematical learning difficulties. However, most current systems focus primarily on practice-based learning, diagnostic assessment, or engagement mechanisms independently. Only a limited number of systems integrate diagnostic screening, adaptive learning pathways, learning analytics, and stakeholder monitoring tools within a single platform. Therefore, there is a clear need for an integrated digital learning intervention system that combines adaptive learning algorithms, multisensory learning activities, gamified educational mechanisms, and continuous performance monitoring to more effectively support children experiencing dyscalculia-related learning difficulties.

1.3 Research Gap

Although significant progress has been made in understanding neurodevelopmental learning disorders such as dyscalculia, existing research and technological solutions still present several limitations in terms of accessibility, personalization, and comprehensive intervention. Many current studies focus on individual aspects such as diagnosis, practice-based learning tools, or engagement mechanisms separately, rather than providing an integrated system that combines early detection, adaptive learning, and continuous progress monitoring within a single platform [15].

Traditional approaches for identifying dyscalculia primarily rely on clinical psychological assessments, standardized diagnostic tests, and specialist educational evaluations. Although these methods provide reliable diagnostic outcomes, they require trained professionals, specialized assessment tools, and considerable financial resources. Consequently, these services remain inaccessible to many families, particularly in regions where educational and psychological support services are limited. The absence of accessible early screening mechanisms often leads to delayed identification of learning difficulties, preventing timely intervention during the critical early stages of cognitive development [16].

Existing digital learning tools developed for mathematics education have attempted to address learning challenges through gamified exercises and interactive activities. However, most of these systems are designed for general mathematics practice rather than targeting the specific cognitive difficulties associated with dyscalculia, such as number sense deficits, magnitude comparison difficulties, and challenges in arithmetic processing. As a result, children with dyscalculia may find it difficult to benefit from these platforms because the learning activities are not tailored to their individual cognitive needs [17].

Recent research has also explored the use of machine learning and learning analytics techniques to detect dyscalculia by analyzing learner performance patterns in mathematical tasks. These approaches demonstrate promising potential for identifying students at risk of mathematical learning difficulties. Nevertheless, many of these systems function primarily as diagnostic or classification tools, focusing on identifying the presence of dyscalculia rather than providing adaptive intervention strategies that support the improvement of mathematical skills [18].

In addition, while gamification has been widely recognized as an effective approach for improving engagement and motivation in digital learning environments, many existing gamified educational applications lack adaptive learning mechanisms capable of dynamically adjusting the difficulty and type of learning activities based on learner performance. Without such personalization, learners may either become discouraged by tasks that are too difficult or disengaged due to repetitive and overly simple exercises [19].

Another limitation observed in existing digital learning platforms is the absence of comprehensive progress monitoring and stakeholder support mechanisms. Many current systems do not provide detailed learning analytics or progress reports that allow parents, teachers, or specialists to track a child's learning development over time. Without continuous monitoring and feedback, it becomes difficult to evaluate the effectiveness of learning interventions or adjust instructional strategies accordingly [20].

Furthermore, most existing digital learning systems rely primarily on symbolic mathematical representation and text-based exercises, which may not effectively support children with

dyscalculia who often require multisensory learning approaches that combine visual, auditory, and interactive learning modalities. The lack of such multisensory learning mechanisms limits the effectiveness of many current platforms in supporting conceptual understanding of numerical concepts.

Therefore, there is a clear need for a comprehensive, intelligent, and accessible digital learning intervention system that integrates multiple functionalities within a single platform. Such a system should combine early screening, personalized learning pathways, adaptive difficulty adjustment, multisensory learning activities, gamified engagement mechanisms, and continuous performance monitoring.

The proposed research addresses this gap by developing a game-based adaptive mobile learning platform designed to support children with dyscalculia. The system integrates diagnostic screening, personalized learning activities, and adaptive learning mechanisms within a unified platform. In addition, the proposed solution incorporates multisensory learning techniques that combine visual object representation, Sinhala audio guidance, and interactive touch-based learning activities to strengthen number recognition and conceptual understanding. Through the integration of adaptive learning algorithms, gamified educational tasks, and analytics dashboards for parents and teachers, the proposed system aims to provide an effective and accessible digital intervention solution for improving numeracy skills among children with dyscalculia.

Feature	[7]	[8]	[9]	[10]	[11]	[12]	[13]	[14]	Proposed System
Adaptive learning mechanism	✗	✗	Limited	✗	✗	✗	Limited	✗	✓
Multi-dimensional learning analysis (number sense, arithmetic accuracy, response time)	✓	✓	✗	✓	✓	✓	Limited	Limited	✓
Game-based learning tasks	✗	✓	✓	✗	✗	✗	✓	✓	✓
Dyscalculia-specific learning intervention	✗	✗	✗	✗	✗	✗	Limited	Limited	✓
Learning analytics and performance evaluation	✗	✗	✗	✗	✓	✓	✗	✗	✓
Parent / teacher progress dashboard	✗	✗	✗	✗	✗	✗	✗	Limited	✓
Continuous progress monitoring	✗	✗	✗	✗	Limited	Limited	✗	✗	✓
Integrated screening and intervention system	✗	✗	✗	✗	✗	✗	✗	✗	✓

Table 01 - System Comparison

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to design and develop an integrated, game-based mobile learning intervention system for children aged 6-8 years to support dyscalculia-related learning difficulties. The system aims to assess foundational numeracy skills, identify learning challenges related to number sense, magnitude comparison, and basic arithmetic reasoning, and provide personalized learning activities through an adaptive learning mechanism.

The main objective of this research is to design and develop an integrated, game-based mobile learning intervention system for children aged 6-8 years to support dyscalculia-related learning difficulties. The system aims to assess foundational numeracy skills, identify learning challenges related to number sense, magnitude comparison, and basic arithmetic reasoning, and provide personalized learning activities through an adaptive learning mechanism with continuous performance monitoring and caregiver-friendly reporting.

2.2 Specific Objectives

1. **Initial Level Identification** - To identify the child's numeracy proficiency level (mild, moderate, or severe) through structured game-based mathematical assessment tasks. The assessment evaluates key indicators such as number recognition, counting ability, magnitude comparison, and basic arithmetic reasoning to establish a baseline numeracy learning profile.
2. **Develop Adaptive Game-Based Learning Activities** - To design and implement engaging, child-friendly digital numeracy learning activities that support the development of key mathematical competencies such as number sense, counting skills, number comparison, place value understanding, and basic arithmetic operations. The system will incorporate **multisensory learning elements**, including visual representations, audio guidance, and interactive game mechanics to improve learner engagement and conceptual understanding.
3. **Personalized Learning Activity Recommendation** - To develop a rule-based scoring mechanism that analyzes learner performance indicators such as answer accuracy, response time, error frequency, and task completion consistency. Based on predefined threshold criteria, the system will recommend appropriate learning activities tailored to the learner's difficulty level and learning progress.
4. **Develop Continuous Performance Tracking and Reporting Dashboard** - To develop a progress monitoring and analytics dashboard that records longitudinal learning data, visualizes improvement trends across multiple learning sessions, and generates simplified progress reports for parents and teachers. The system will highlight improvements in numerical understanding, arithmetic accuracy, and response speed to support effective learning intervention.

3. METHODOLOGY

Research approach

This research adopts the Design Science Research (DSR) methodology combined with an experimental evaluation approach to design, develop, and validate a digital learning intervention system for children with developmental dyscalculia.

The research process follows several stages including problem identification, requirement analysis, system design, system development, and experimental evaluation. Initially, the study analyzes the challenges faced by children with dyscalculia and examines the limitations of existing learning tools. Based on these findings, the research proposes a game-based adaptive mobile learning system designed to assess numeracy difficulties and provide personalized intervention for children aged 6-8 years.

The proposed system integrates several components including game-based numeracy assessment tasks, rule-based performance evaluation, adaptive learning activity recommendation, and progress monitoring dashboards for parents and teachers.

Children's mathematical abilities are evaluated through interactive learning tasks that measure indicators such as accuracy, response time, task completion rate, and error patterns. Based on these indicators, a rule-based algorithm classifies the learner's difficulty level into predefined categories such as mild, moderate, or severe, and recommends targeted learning activities to improve numeracy skills.

During system usage, learner performance data is continuously collected and analyzed. The system adjusts task difficulty dynamically using an adaptive learning mechanism, ensuring that learning activities remain appropriately challenging while maintaining learner engagement.

The system development process follows an Agile iterative software development model, allowing continuous improvement through repeated cycles of design, implementation, testing, and refinement.

Finally, an experimental evaluation will be conducted by comparing baseline numeracy performance with post-intervention outcomes to measure improvements in accuracy, response time, and task completion rates. This evaluation helps validate the effectiveness of the proposed system in supporting children with dyscalculia.

Technical methods and algorithms

Numerical Interaction Capture

Numerical interaction capture forms the foundation of data collection, where every response generated by the child during interactive learning activities is recorded as structured performance data. This includes numerical selections, drag-and-drop actions, counting interactions, and arithmetic responses captured through touchscreen events.

Each interaction is logged with associated timestamps, response accuracy, and completion duration. This raw interaction data enables detailed analysis of how numerical reasoning tasks are performed rather than simply determining whether answers are correct or incorrect.

Number Sense Analysis

Number sense analysis evaluates a child's ability to recognize numerical magnitudes, compare quantities, and understand number ordering. Performance indicators are calculated using response accuracy, decision time, and consistency across multiple tasks. Statistical analysis of response times and accuracy scores helps identify patterns that indicate potential number sense deficits associated with dyscalculia.

Magnitude Comparison Detection

Magnitude comparison detection examines the learner's ability to determine relative values between numbers or quantities. In tasks where children must select the larger or smaller number, the system records response correctness, hesitation duration, and repeated error patterns. These indicators help identify conceptual misunderstandings in numerical magnitude representation.

Counting Ability Assessment

Counting ability assessment evaluates sequential number recognition and counting accuracy using structured game-based activities. The system analyzes whether children correctly count objects, identify missing numbers in sequences, or complete number patterns. Errors such as skipped numbers, incorrect sequencing, or repeated counting are captured and categorized using rule-based error classification techniques.

Arithmetic Reasoning Analysis

Arithmetic reasoning analysis focuses on basic mathematical operations such as addition and subtraction within developmentally appropriate number ranges. The system evaluates arithmetic accuracy, solution time, and problem-solving consistency. Incorrect answers are further analyzed to determine whether errors are random or represent systematic misconceptions such as misunderstanding place value or misinterpreting arithmetic operations.

For activities requiring verification of arithmetic results, the system compares the child's response with predefined correct solutions stored in the system database and provides immediate feedback to guide learning.

Dyscalculia Severity Classification

A weighted scoring algorithm integrates multiple performance indicators to classify dyscalculia severity levels (mild, moderate, or severe). The scoring model combines parent questionnaire responses—which capture observational insights regarding the child's numerical learning behavior—with directly measured performance metrics obtained from interactive assessment tasks.

These metrics include numerical accuracy, response time variability, task completion rates, and error frequency.

Adaptive Learning Recommendation Engine

The adaptive learning engine operates using a rule-based decision matrix that considers the learner's identified severity level, error patterns, and ongoing performance trends. Based on this analysis, the system dynamically selects appropriate learning games and adjusts task difficulty to provide targeted support for improving number sense, counting skills, and arithmetic reasoning.

Tools and platforms with justification

The frontend application is developed using React Native, selected for its component-based architecture that supports reusable user interface elements across multiple learning modules and enables efficient rendering of interactive mathematical activities. React Native allows the development of visually engaging educational games with smooth animations and real-time feedback, which are essential for maintaining motivation and attention among young learners.

Interactive learning activities are implemented using HTML5 Canvas integrated with React-Konva, which provides a lightweight and high-performance graphical rendering environment. This framework enables dynamic visualization of mathematical objects such as numbers, counters, shapes, and drag-and-drop elements used in numeracy learning tasks. It ensures smooth interaction while capturing detailed user input events necessary for performance analysis.

The backend architecture utilizes Node.js with Express, chosen for its event-driven and non-blocking architecture that efficiently manages concurrent user sessions and real-time data processing. Using JavaScript for both frontend and backend development also simplifies system integration and maintenance.

A dual-database architecture is adopted to balance flexibility and real-time performance:

- **MongoDB** stores user profiles, learning progress records, game configurations, and aggregated performance analytics. Its document-based schema allows flexible storage of structured learning data.
- **Firebase** supports real-time synchronization of active learning sessions and performance updates. Firebase Authentication is used for secure user management and access control.

This combined architecture ensures scalability, efficient data processing, and real-time feedback, while remaining cost-effective for educational deployments.

Version control is managed using Git, with the project repository hosted on GitHub to support collaborative development and version tracking. The development environment utilizes Visual Studio Code (VS Code) as the primary integrated development environment (IDE) for coding, debugging, and system management.

Data collection and ethical considerations

The data collection process focuses on gathering numeracy learning performance data and expert educational insights related to dyscalculia characteristics rather than collecting sensitive personal information from children. This approach minimizes ethical concerns while maintaining research validity. The system collects learning indicators such as number recognition accuracy, magnitude comparison performance, counting consistency, arithmetic problem-solving accuracy, and response time patterns.

To design assessment tasks, parent questionnaires, and adaptive learning activities, consultations were conducted with primary school mathematics teachers and educational psychologists to identify key numeracy difficulties associated with dyscalculia and appropriate intervention strategies. Their input helped define assessment tasks, difficulty thresholds, and learning activity structures suitable for children aged 6-8 years.

All collected performance data will be anonymized and used solely for research and system evaluation purposes. Participation in system validation will require parental consent, and no personally identifiable information will be stored within the system database.

Validation strategy and performance metrics

Following system development, validation will be conducted with a small group of approximately 5-6 children to evaluate technical performance, usability, and intervention effectiveness.

Technical validation will examine the accuracy of interaction data capture, performance metric calculations, scoring algorithms, and adaptive learning recommendations. System-generated assessments will be compared with expert evaluations to determine classification accuracy.

Pre-intervention and post-intervention comparisons will measure improvements in key numeracy indicators such as:

- Number recognition accuracy
- Magnitude comparison performance
- Counting consistency
- Arithmetic problem-solving speed

Usability evaluation will include child engagement observation, parent feedback surveys, and System Usability Scale (SUS) assessments. Engagement indicators such as session duration, activity completion rates, and repeated usage patterns will also be analyzed.

The target performance metrics include achieving classification accuracy above 80%, strong agreement with expert evaluations, measurable improvement in numeracy skills, and high levels of learner engagement throughout the intervention period.

4.HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed solution is an integrated, game-based adaptive learning platform designed to support children aged 6–8 years who experience dyscalculia-related numeracy difficulties. The system assesses foundational mathematical abilities and provides personalized learning interventions through interactive educational games. It incorporates multisensory learning elements including visual object-based representations, Sinhala audio guidance, and touch-based interaction to improve number recognition, counting ability, and arithmetic reasoning. The platform also includes caregiver-friendly progress dashboards that allow parents and teachers to monitor the child's learning progress and performance over time.

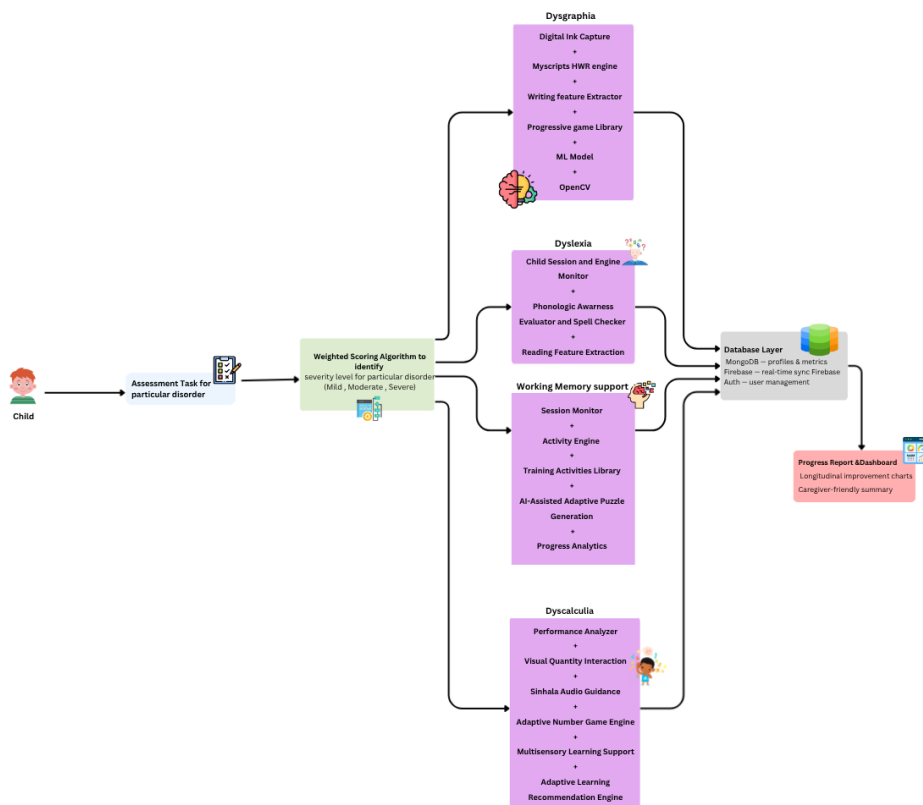


Image 01 - System Diagram

The system begins with the child completing interactive game-based numeracy assessment tasks through the mobile application. These tasks evaluate essential mathematical skills such as number recognition, counting accuracy, magnitude comparison, and simple arithmetic reasoning. During these activities, the system captures performance data including response accuracy, response time, error patterns, and task completion rates. A performance data analyzer processes these indicators and feeds them into a weighted scoring algorithm that classifies the learner's difficulty level into categories such as mild, moderate, or severe.

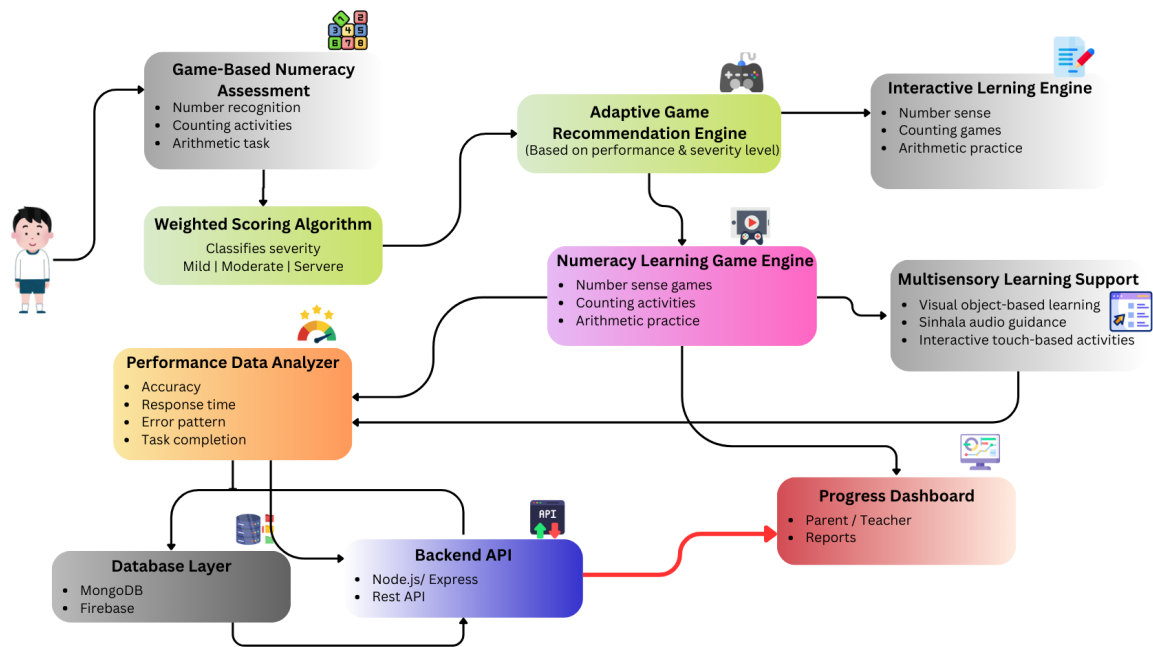


Image 02 - Component diagram for dyscalculia intervention module

Based on the identified learning level and performance trends, the adaptive learning recommendation engine dynamically selects appropriate numeracy learning activities from the game library. These activities are delivered through a numeracy learning game engine that integrates multisensory learning support, combining visual representations of numbers, Sinhala audio instructions, and interactive touch-based tasks to strengthen conceptual understanding of numerical relationships. As the child interacts with the system, new performance data is continuously captured and analyzed, enabling the system to adapt task difficulty and personalize the learning pathway.

Based on the identified learning level and performance trends, the adaptive learning recommendation engine dynamically selects appropriate numeracy learning activities from the game library. These activities are delivered through a numeracy learning game engine that integrates multisensory learning support, combining visual representations of numbers, Sinhala audio instructions, and interactive touch-based tasks to strengthen conceptual understanding of numerical relationships. As the child interacts with the system, new performance data is continuously captured and analyzed, enabling the system to adapt task difficulty and personalize the learning pathway.

The system architecture is designed with scalability and security in mind. The use of a modular architecture with React Native for the frontend and Node.js with Express for backend services enables efficient handling of multiple concurrent users and supports the future integration of additional learning modules. Data security is maintained through secure authentication mechanisms, encrypted communication protocols, and controlled API access to protect sensitive learning data. This scalable and secure architecture ensures that the platform can support a growing number of learners while maintaining reliable performance and data protection.

5.USER REQUIREMENTS

Stakeholder identification

Primary Users: Children aged 6-8 years who experience mathematical learning difficulties associated with dyscalculia, including challenges with number recognition, counting, magnitude comparison, and basic arithmetic reasoning.

Secondary Users: Parents, teachers, or guardians who are responsible for creating and managing child profiles, monitoring progress reports, and supervising system usage during learning sessions. They can review performance summaries and track improvements in the child's numeracy learning progress.

Tertiary Users: Educational specialists and learning support professionals who may provide expert input for assessment design, validate system classifications, and use system-generated reports to support educational interventions. School educators may also use these reports to provide additional classroom support for children experiencing numeracy learning difficulties.

Functional requirements

01. **User Registration and Login** - The system shall allow guardians to register and securely log in using email and password authentication to access the platform and manage child profiles.
02. **Child Profile Setup** - Guardians shall be able to create and manage child profiles including information such as name, age, and gender to enable personalized numeracy assessment and intervention.
03. **Game-Based Numeracy Assessment** - The system shall provide interactive, gamified numeracy assessment tasks that evaluate key mathematical skills including number recognition, counting accuracy, number ordering, magnitude comparison, and basic arithmetic reasoning.
04. **Dyscalculia Level Classification** - The system shall analyze performance metrics extracted from assessment tasks using a weighted scoring algorithm to classify children into mild, moderate, or severe dyscalculia severity levels.
05. **Adaptive Game Recommendation** - The system shall recommend appropriate numeracy intervention games based on the child's identified difficulty level and performance patterns.
06. **Interaction Data Capture** - The system shall record all child interactions during learning activities including responses, response time, task completion rate, and error patterns to preserve detailed learning behavior data.

07. **Answer Correctness Verification** - The system shall verify the correctness of the child's responses by comparing submitted answers with predefined correct solutions stored in the system database.
08. **Progressive Game Library** - The system shall maintain a structured library of numeracy learning games with adjustable difficulty parameters including number ranges, visual aids, arithmetic complexity, and response time limits.
09. **Real-Time Performance Tracking** - The system shall log child performance during game activities including accuracy, completion time, error frequency, and engagement indicators to enable continuous learning analysis.
10. **Caregiver-Friendly Report Generation** - The system shall generate comprehensive progress reports for guardians summarizing the child's numeracy performance, identified difficulties, severity level, and improvement trends over time.

Non-functional requirements

1. **Performance** - The system shall capture and process child interactions in real time with minimal latency to ensure a smooth and responsive learning experience. Performance metrics and activity results shall be generated within four seconds after task completion. Game animations and transitions shall maintain smooth performance at a minimum of 30 frames per second to sustain engagement for young learners.
2. **Reliability** - Learning data shall be logged consistently across sessions. The system shall support offline data caching and automatically synchronize data with the server once internet connectivity is restored.
3. **Scalability** - The database architecture shall support scalability to store data for at least 1,000 child profiles without performance degradation. The system shall handle increasing numbers of concurrent learning sessions using cloud-based scalable infrastructure capable of scaling on demand.
4. **Usability** - The child interface shall incorporate bright colors, friendly animations, and age-appropriate visual elements to maintain engagement for children aged 6-8 years. The system shall provide multisensory feedback including visual rewards (stars and animations) and audio responses upon task completion to reinforce positive engagement. The guardian interface shall present results using clear, non-technical language and visual progress indicators to support easy understanding.
5. **Security and Privacy** - Child learning data shall be securely stored, encrypted, and anonymized to protect user privacy. Access to reports shall be restricted to authorized

guardians, teachers, and approved professionals through secure authentication mechanisms.

6. **Maintainability** - The system architecture shall follow a modular design with clear separation between the frontend, backend, and database layers to support independent updates and maintenance. The codebase shall be well-documented and adhere to standard development practices to ensure maintainability. Additionally, new learning games and assessment tasks shall be added to the game library without modifying the core system architecture.

6. COMMERCIALIZATION PLAN

Target market and customer persona

The primary target market for the proposed dyscalculia intervention system is the education and child development sector, particularly supporting children aged 6-8 years who experience mathematical learning difficulties. Educational institutions such as primary schools, special education centers, tutoring institutes, and child development clinics can benefit from structured digital tools designed to support early numeracy development. In addition, parents seeking accessible home-based learning support represent an important user group. In many regions, limited access to educational psychologists and specialized learning interventions makes support difficult, highlighting the need for scalable digital learning platforms that can provide accessible and effective numeracy support.

User Personas:

Parents and Guardians - Individuals seeking accessible digital tools to support their child's numeracy learning at home.

Primary School Educators - Teachers who require structured digital resources to assist students experiencing difficulties with foundational numeracy skills.

Learning Support Professionals - Educational therapists, special education teachers, and learning specialists who require monitoring tools to track student progress and support intervention planning.

1. Child User Persona

Name	Nethmi Perera
Age	7 years
Background	Nethmi enjoys playing games on tablets but finds mathematics activities confusing and frustrating. She often struggles with recognizing numbers, counting objects correctly, and solving simple arithmetic problems in class.
Goals	Understand numbers and counting clearly. Solve simple math problems with confidence. Enjoy learning mathematics through games.
Challenges	Difficulty recognizing number relationships and quantities. Slow response when solving math questions. Low confidence in mathematics compared to other subjects.
How the System Helps	The application provides game-based numeracy activities, visual objects, and audio instructions to help Nethmi practice counting, number recognition, and simple arithmetic in an engaging and supportive way.

2. Therapist Persona

Name	Mrs. Perera
Age	40 years
Background	Mrs. Perera is concerned about her child's difficulty in mathematics but is unsure how to provide proper learning support at home.
Goals	Help her child improve numeracy skills. Track the child's learning progress. Find an easy-to-use learning tool for home practice.
Challenges	Limited knowledge of how to support math learning difficulties. Difficulty identifying the child's weak areas in numeracy. Limited access to specialized learning support.
How the System Helps	The system provides progress reports and performance insights, helping parents understand their child's learning level and support practice at home.

Value proposition

Value Element	Description
Affordable Intervention	Reduces reliance on expensive tutoring or therapy sessions by providing an accessible digital learning platform for daily numeracy practice at a significantly lower cost.
Clear Progress Visibility	Caregiver-friendly reports provide clear insights into improvements in number recognition, counting ability, magnitude comparison, and arithmetic accuracy over time.
Inclusive Classroom Support	Enables teachers to support students with mathematical learning difficulties without requiring specialized training in learning disorders.
Time Savings	Automates numeracy assessment and progress tracking, reducing the time required for teachers and parents to monitor student performance.
Engaging Game Experience	Mathematical learning activities are delivered through interactive games, animations, and reward mechanisms to maintain high levels of motivation and engagement among young learners.
Multisensory Learning Approach	Combines visual object representation, audio guidance, and interactive activities to strengthen number sense and improve conceptual understanding for children with dyscalculia.
Builds Mathematical Confidence	Positive reinforcement and visible progress help reduce math-related anxiety and improve children's confidence in solving numerical problems.
Age-Appropriate Design	Bright visuals, simple interfaces, friendly characters, and audio guidance are designed specifically for children aged 6-8 years.
Accessible Home Learning Support	Provides structured numeracy learning support that can be used both at home and in classroom environments.

Table 02 - value proposition

Revenue model

To ensure both accessibility and long-term sustainability, the proposed dyscalculia intervention system adopts a hybrid revenue model that combines subscription-based access, institutional licensing, freemium services, and strategic partnerships. This approach enables wide accessibility for families while supporting large-scale adoption within educational institutions.

1. Subscription Model

Parents and guardians can subscribe to the platform on a monthly or yearly basis to access personalized numeracy learning activities and progress monitoring tools.

Multiple subscription tiers such as Basic, Standard, and Premium can be offered based on the level of features available. Premium plans may include advanced functionalities such as adaptive learning modules, detailed performance analytics, and extended learning game libraries.

2. Institutional Licensing

Educational institutions including primary schools, tutoring centers, and special education support centers can purchase annual institutional licenses that allow multiple student accounts under a single organizational subscription. This model enables teachers and learning support professionals to monitor student progress, analyze learning outcomes, and implement targeted numeracy interventions within classroom environments.

3. Freemium Model

A freemium access model will be implemented to encourage widespread adoption. The free version of the system will provide access to basic numeracy assessment tasks and a limited set of learning activities. Advanced features such as adaptive learning pathways, multisensory learning modules, and detailed progress reports will be available through paid subscription plans.

4. B2B Partnerships

Strategic partnerships can be established with educational institutions, child development centers, pediatric clinics, and learning support organizations. Through these partnerships, the system can be integrated into existing child development programs and early learning support services, expanding the platform's reach while creating additional revenue opportunities.

Cost estimation

Cost Category	Estimated Cost (LKR)	Details
Software Development	Rs. 35,000 - 50,000	React Native frontend, Node.js backend, database integration.
Game Design	Rs. 10,000 - 15,000	Child-friendly animations, characters, UI elements for game activities
Expert Consultation	Rs. 10,000 - 15,000	Compensation for occupational therapists and pediatric specialists for validation
MyScript Integration	Rs. 0 - 10,000	Free tier available, minimal during development
Testing and Piloting	Rs. 10,000 - 15,000	Recruitment of participants, incentives, usability testing.
TOTAL INITIAL COSTS	Rs. 65,000 - 100,000	

Table 03 - Cost Estimation

Pricing strategy

A **penetration pricing strategy** will be adopted during the initial launch phase to encourage early adoption among parents, schools, and learning support institutions. This strategy allows the platform to quickly reach a wider user base while demonstrating its effectiveness in improving numeracy skills among children with dyscalculia.

- **Basic Plan:** A low-cost monthly subscription that provides access to basic numeracy assessment activities and a limited set of interactive learning games.
- **Standard Plan:** Provides full access to adaptive game-based learning activities, multisensory numeracy modules (visual, audio, and interactive tasks), personalized learning recommendations, and detailed progress monitoring reports.
- **Institutional Plan:** Offers discounted bulk pricing for schools, tutoring centers, and learning support programs that require multiple student accounts and centralized progress monitoring tools for teachers.

The pricing model will remain affordable and accessible, particularly for families and educational institutions in resource-constrained environments, while ensuring financial sustainability for platform development, system maintenance, and future feature expansion.

Competitive advantage

The proposed system offers several competitive advantages compared to existing numeracy learning applications. It is specifically designed for Sinhala-language learners, making the platform more accessible and suitable for children in local educational contexts. In addition, the system utilizes a rule-based and interpretable scoring mechanism that explains the learner's numeracy difficulty level using measurable indicators such as accuracy, response time, and error patterns. This transparency allows parents and teachers to clearly understand the child's learning progress and identify potential numeracy difficulties.

Second, the system incorporates a closed-loop adaptive learning mechanism that dynamically adjusts the difficulty level of learning activities based on the child's ongoing performance. This adaptive approach ensures that learners receive appropriately challenging tasks while reducing frustration and maintaining engagement during learning activities.

Third, the platform integrates gamified multisensory learning activities designed for children aged 6-8 years. These activities combine visual objects, audio instructions, and interactive touch-based tasks to improve engagement and strengthen conceptual understanding of mathematical skills such as number recognition, counting, magnitude comparison, and basic arithmetic reasoning.

Finally, the system performs real-time numeracy performance analysis and provides continuous progress monitoring. Unlike most existing numeracy learning applications that mainly focus on practice-based exercises, the proposed system integrates structured assessment, adaptive intervention, and learning analytics within a single platform to provide more effective support for children experiencing dyscalculia-related learning difficulties.

Intellectual property considerations

The project may involve protecting the software design, algorithms, and educational content used in the platform. The system can be protected through copyright registration for the software code and educational game content. If the handwriting analysis algorithms or adaptive recommendation methods demonstrate novelty, they may also be considered for patent protection. Furthermore, the application name and logo can be protected through trademark registration to support future commercialization.

7.BUDGET AND JUSTIFICATION

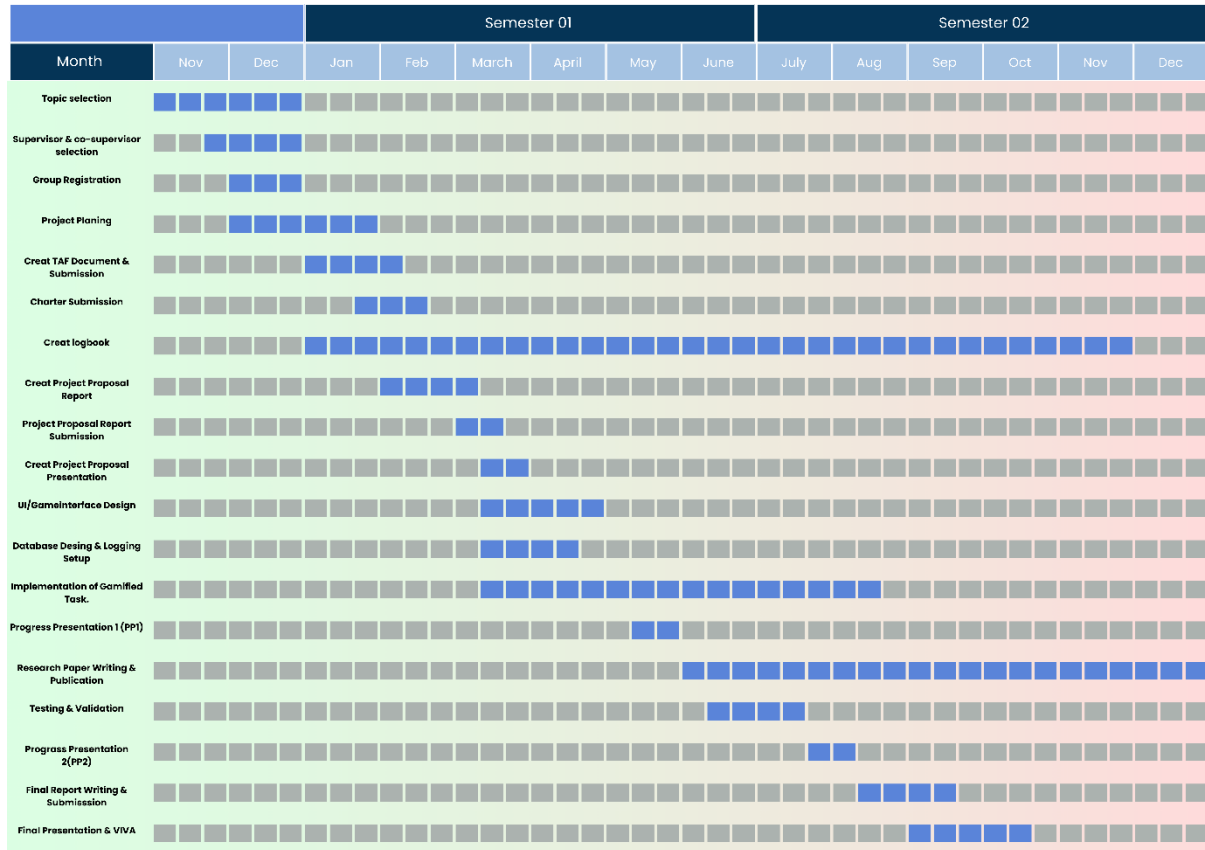
Cost Category	Estimated Cost (LKR)	JUSTIFICATION
Travel expenses	9,000	Visits to the SLIIT Matara Center to meet the Supervisor and Co-supervisor. Visit to Ayati Center and Other Disability Treatment Centers.
Data collection	10,000	Arrange private appointments to collect data and expert insights regarding dysgraphia.
Printing & documentations	1,500	Printing proposals, reports, consent sheets and questionnaires.
Total	20,500	

Table 04 - Budget and Justification

7.WORK BREAKDOWN STRUCTURE (WBS)

Phase	Task	Timeline
Project Planning & Requirement Analysis	Literature review and problem identification	Week 1 - Week 4
	Define system requirements and functional specifications	Week 5 - Week 6
	Design research methodology and technical approach	Week 7 - Week 8
	Prepare project proposal documentation	Week 9- week10
System Design	System architecture and workflow design	Week 10 - Week 11
	Database schema design (MongoDB, Firebase)	Week 12
	UI/UX design for learning games and dashboards	Week 13 - Week 14
Frontend Development	Develop login, guardian profile, and questionnaire modules	Week 15 - Week 17
	Implement interactive numeracy learning interface.	Week 19 - Week 20
	Develop game-based numeracy learning activities.	Week 21 - Week 23
	Implement audio instructions and multisensory interaction features.	Week 24
Backend Development	Implement API endpoints and database integration	Week 26 - Week 27
	Implement rule-based scoring algorithm and dyscalculia classification	Week 28
Advanced Features	Implement performance analytics and learning evaluation module.	Week 29
	Develop continuous progress monitoring system.	Week 30 - Week 31
	Develop progress monitoring dashboards	Week 32
System Testing & Validation	Functional testing of modules	Week 33
	Conduct usability testing and analyze performance results	Week 33 - Week 34
Documentation & Finalization	Prepare final research report, system documentation, and presentation materials	Week 35 - week 36

8. GANTT CHART



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11. APPENDICES

Survey instruments

Smart Learn: An Adaptive Mobile Platform for Neurodevelopmental Learning Disorders

Introduction

Smart Learn: An Adaptive Mobile Platform for Neurodevelopmental Learning Disorders is a research and development project conducted by final-year undergraduate students in Information Technology. The project focuses on designing a supportive digital learning application for children with neurodevelopmental learning difficulties such as dyslexia, dyscalculia, dysgraphia, and working memory challenges.

The purpose of this questionnaire is to gather professional insights from therapists, educators, and specialists about current therapy activities, assessment methods, and effective learning strategies for children aged approximately 6-8 years.

The information collected will be used only for academic research and application development purposes. It will help us design a more effective, adaptive, and child-friendly learning application. All responses will be kept confidential and will only be used to improve the quality, usability, and effectiveness of the proposed mobile platform.

Questions

1. What types of activities or exercises are currently used in therapy for children with dyslexia, dyscalculia, dysgraphia, and working memory difficulties?
2. Are these activities usually paper-based, game-based, digital, or therapist-guided?
3. Which activities do you find most effective for children aged 6-8?
4. How do these activities change as the child improves or struggles?
5. Is there any recognized way to categorize these disorders into levels such as mild, moderate, or severe in clinical practice?
6. Do therapists assess children based on functional ability levels rather than biological markers?
7. How do you typically identify a child's current level — through standardized tests, observation, questionnaires, or activities?
8. If an app were to adjust its difficulty level based on a child's performance, what factors should it consider to avoid frustration or overstimulation?
9. What signs indicate a child should move to a higher or lower level?
10. Are there any activities or approaches that should be avoided for children with these disorders?
11. What are the signs that a child is overwhelmed rather than challenged?
12. What kind of progress information is most useful for parents?
13. How do therapists usually communicate improvement to parents?
14. From your professional experience, what is one feature you wish learning apps for neurodevelopmental disorders did better?

Risk analysis

A risk analysis was conducted to identify potential technical and operational risks associated with the dyscalculia learning intervention system. The analysis focuses on core system components such as the numeracy assessment module, adaptive scoring algorithm, gamified learning activities, and backend services.

Risk Factor	Impact	Probability	Mitigation Strategy
Incorrect Numeracy Skill Assessment	High	Medium	Use multiple assessment tasks (number recognition, counting, simple arithmetic) to evaluate the child's performance. The scoring algorithm will combine several metrics to improve accuracy.
Incorrect Severity Classification (Mild, Moderate, Severe)	High	Low	Define threshold values based on expert consultation and test the algorithm using sample datasets to validate classification accuracy.
Children Misinterpreting Game Instructions	Medium	Medium	Use clear visual instructions, animations, and audio guidance to help children understand the game tasks correctly.
Touchscreen Interaction Errors	Medium	High	Implement input validation and confirmation mechanisms to reduce accidental responses and ensure reliable data capture.
Data Privacy and Security Risks	Critical	Low	Use secure authentication, encrypted communication (HTTPS), and anonymized data storage to protect children's information.
Internet Connectivity Issues	Medium	Medium	Implement offline caching for game activities and synchronize data with the server once internet connectivity is restored.

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